

Aliah University

End-Semester Examination (Even Semester) - 2025

(BCA 2ND Year 4TH Semester)

Subject Name: Computer Networks

Full Marks: 80

Subject Code: CSAUGPC18

Time: 3hrs

Group-A

(Answer all questions)

(10 × 1 = 10)

1. Do as Directed Marks :

- a) A _____ is a computer network that covers a large geographic area such as a city, country. [CO1, BL2]
- b) Switch resides in Data Link Layer. (T/F) [CO2, BL2]
- c) IP address 122.168.10.1 belongs to class A [CO5, BL3]
- d) IP address is 32 bit long. (True/False) [CO5, BL3]
- e) CSMA/CA stands for _____ [CO3, BL2]
- f) The Internet model consists of four layers. [CO3, BL2]
- g) What is the main function of data link layer? [CO2, BL2]
- h) What is point to point connection? [CO2, BL2]
- i) TCP/IP is based on a seven-layer reference model. (True/False) [CO1, BL1]
- j) A bit string 011110111110111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing? [CO2, BL5]

Group-B

(Answer any five questions)

(5*6= 30)

- 2. What is Nyquist Bit Rate? Consider a noiseless channel with a bandwidth of 6000 hz transmitting a signal with two signal levels. Find bit rate. (2+4=6) [CO2, BL5]
- 3. Define Bit Rate and baud rate. An analog signal carries 5 bits in each signal unit. If 1200 signal units are sent per second, find the baud rate and bit rate. (3+3=6) [CO2, BL4]
- 4. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28.

- ✓ a. What is the first address in the block?
- ✓ b. Find the last address for the block
- ✓ c. Find the number of addresses. $(2+2+2=6)$ [CO5, BL5]

- ✓ 5. Explain OSI model in networking. (6) [CO1, BL2]
- 6. State the advantages of CSMA/CA. Explain the token passing algorithm with neat diagram. $(1+5=6)$ [CO4, BL3]
- ✓ 7. Explain Selective Repeat ARQ with diagram. (6) [CO4, BL4]

Group-C

(Answer any four questions)

$(4 \times 10 = 40)$

- ✓ 8. Explain interior and exterior routing with diagram. List the different types of routing algorithm and explain any one of them. $(4+6=10)$ [CO4, BL2]
- ✓ 9. Write down the disadvantages of simple and two dimensional parity check method. Given a 10bit sequence 101001111 and a divisor of 1011. Find the CRC. $(4+6=10)$ [CO3, BL5]
- ✓ 10. ~~Explain comparison between OSI and TCP/IP model. Compare between Packet Switching and Message switching. $(5+5=10)$ [CO2, BL2]~~
- ✓ 11. Define IP address and Default mask. Compare IPv4 vs IPv6. Explain subnetting with diagram. $(4+2+4=10)$ [CO4, BL5]
- 12. Write short notes: a) Multiplexing b) TCP c) Spread Spectrum. $(3+3+4=10)$ [CO5, BL3]